FAST FOOD ANALYSIS - TACO BELL

Project Title: **CDACL-004-Fast Food Analysis**

Tools Used: SQL, Microsoft Excel, Power BI

Dataset: project_food

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Introduction

This project analyses fast-food nutritional data, with a specific focus on Taco Bell. The objective is to evaluate the nutritional characteristics of menu items, identify high- and low-nutrient products, and understand the distribution of calories, fats, sodium, sugar, protein, and carbohydrates.

Fast-food restaurants offer a wide range of menu items with different nutritional compositions. Analysing these items helps identify products with high calorie, fat, sugar, or sodium content and compare their nutritional characteristics.

SQL was used to query and analyse the dataset, Excel was used for descriptive statistics and nutritional comparisons, and Power BI was used to create an interactive dashboard.

The findings are intended to support menu evaluation, nutritional awareness, and data-driven decision-making.

Objectives

- Analyse Taco Bell menu items using the available nutritional attributes.
- Calculate average and overall nutritional statistics.
- Identify items with the highest and lowest calories.
- Analyse fat, saturated fat, trans fat, sodium, sugar, protein, and carbohydrates.
- Compare Taco Bell's nutritional profile with other restaurants in the dataset.
- Develop a Power BI dashboard to communicate important findings.
- Provide recommendations based on the analysis.

Dataset Description

The dataset contains fast-food menu information from multiple restaurant chains.

The main attributes include restaurant, item, calories, calories from fat, total fat, saturated fat, trans fat, cholesterol, sodium, total carbohydrates, fiber, sugar, protein, vitamins and calcium.

The analysis uses the available dataset and focuses on Taco Bell for the detailed company level investigation.

Tools Used:

SQL: Used to inspect the dataset, examine records, calculate aggregates, and identify nutritional extremes.

Microsoft Excel: Used to calculate descriptive statistics, including averages and other summary measures, and organize the results.

Power BI: Used to transform data, build visualizations, compare nutritional characteristics, and present the findings in a dashboard.

Data Preparation

The data was initially retrieved using MySQL. The Food table was used for analysis. Data was used to Checked the table structure and data quality and Analysed restaurant and menu item records. Required fields were transformed into appropriate formats for analysis.

Data was examined missing values, duplicate menu items, and numerical attributes and Calculated nutritional statistics in Excel.

Power BI was used for Data Transformation and to create two dashboard pages.

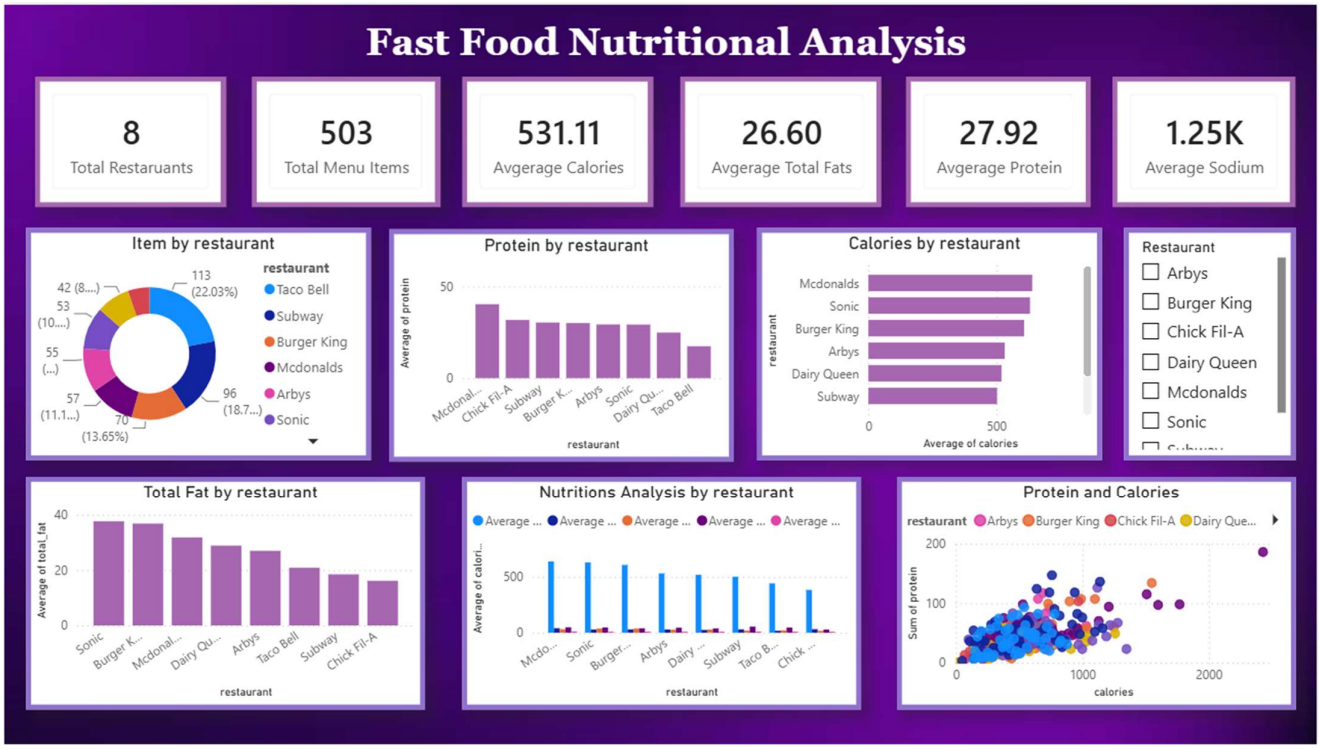
Data Transformation:-

- Verified column data types
- Checked data quality
- Checked for errors
- Verified that there were no NULL values in the analysed fields
- Prepared the data for visualization

Dashboard Analysis

The Power BI dashboard presents nutritional information in a visual format. It allows users to compare menu items and understand differences in their nutritional composition.

The dashboard includes the visualizations developed during the project, such as nutritional comparisons by restaurant and the selected food-item analyses.



Taco Bell Menu Analysis:

Taco Bell was selected for detailed analysis. Its menu items were evaluated using the available nutritional attributes.

The analysis examined:

- Average calories and other nutritional values.
- Highest-calorie and lowest-calorie items.
- Items with high total fat and saturated fat.
- Items with high sodium and sugar.
- Items with high protein.
- Nutritional differences between menu items.



Overall Nutritional Analysis:

The overall dataset was Analysed to understand the distribution of calories, fat, sodium, sugar, and protein across menu items.

The analysis included average nutritional values and comparisons between restaurant chains. These results provide a general overview of nutritional differences in the dataset.

Row Labels	Average of total_fat	Average of calcium	Average of protein	Average of sugar	Average of fiber	Average of total_carb	Average of sat_fat	Average of cholesterol	Average of calories
Arbys	26.98181818	17.36	29.25454545	7.563636364	2.709090909	44.87272727	7.972727273	70.45454545	532.7272727
Burger King	36.81428571		30.01449275	8.185714286	2.383333333	39.31428571	11.15	100.8571429	608.5714286
Chick Fil-A	16.14814815	11.32	31.7037037	4.148148148	2.32	28.62962963	4.111111111	79.07407407	384.4444444
Dairy Queen	28.85714286	16.40740741	24.83333333	6.357142857	2.833333333	38.69047619	10.44047619	71.54761905	520.2380952
Mcdolds	31.80701754	20.59649123	40.29824561	11.07017544	3.228070175	48.78947368	8.289473684	109.7368421	640.3508772
Sonic	37.64150943	17.24489796	29.18867925	6.528301887	2.660377358	47.20754717	11.41509434	86.98113208	631.6981132
Subway	18.47916667	39.125	30.3125	10.09375	6.5625	54.71875	6.197916667	61.30208333	503.0208333
Taco Bell	20.89565217	24.80769231	17.4173913	3.704347826	5.713043478	46.63478261	6.591304348	39.04347826	443.6521739
(blank)									
Grand Total	**26.59029126**	**24.85245902**	**27.89105058**	**7.262135922**	**4.137176938**	**45.66407767**	**8.153398058**	**72.45631068**	**530.9126214**

SQL Analysis

SQL was used to explore, validate, and Analyse the fast-food dataset. The analysis focused on restaurant-wise menu counts, data quality, nutritional averages, and the identification of menu items with extreme nutritional values.

The database used for the project was project_food, and the table was food.

Database and Table Inspection:

The database and table structure were inspected to understand the available attributes and verify that the data was loaded.

```
SHOW DATABASES;
USE project_food;

SHOW TABLES;
DESCRIBE food;
```

Restaurant-Wise Menu Item Count:

To compare the number of menu-item records available for each restaurant.

Taco Bell had the largest number of records in the dataset, with 115 items, followed by Subway with 96 and Burger King with 70, based on the previously obtained results.

```
SELECT restaurant, COUNT(*) AS item_count
FROM food
GROUP BY restaurant
ORDER BY item_count DESC;

SELECT COUNT(*) AS total_rows,
       COUNT(DISTINCT restaurant) AS total_restaurants,
       COUNT(DISTINCT item) AS total_items
FROM food;
```

Average Nutritional Values by Restaurant:

To calculate average nutritional values for each restaurant and compare their nutritional profiles.

```sql
SELECT
    restaurant,
    COUNT(*) AS total_items,
    ROUND(AVG(calories), 2) AS avg_calories,
    ROUND(AVG(cal_fat), 2) AS avg_calories_from_fat,
    ROUND(AVG(total_fat), 2) AS avg_total_fat,
    ROUND(AVG(sat_fat), 2) AS avg_saturated_fat,
    ROUND(AVG(trans_fat), 2) AS avg_trans_fat,
    ROUND(AVG(cholesterol), 2) AS avg_cholesterol,
    ROUND(AVG(sodium), 2) AS avg_sodium,
    ROUND(AVG(total_carb), 2) AS avg_carbs,
    ROUND(AVG(fiber), 2) AS avg_fiber,
    ROUND(AVG(sugar), 2) AS avg_sugar,
    ROUND(AVG(protein), 2) AS avg_protein
FROM food
GROUP BY restaurant
ORDER BY avg_calories DESC;
```

Taco Bell Nutritional Analysis:

To summarize the nutritional profile of Taco Bell menu items. The output provides the average nutritional values for Taco Bell and serves as the basis for the company-specific dashboard and diagnostic analysis.

```sql
SELECT
    COUNT(*) AS total_items,
    ROUND(AVG(calories),2) AS avg_calories,
    ROUND(AVG(cal_fat),2) AS avg_cal_fat,
    ROUND(AVG(total_fat),2) AS avg_total_fat,
    ROUND(AVG(sat_fat),2) AS avg_sat_fat,
    ROUND(AVG(trans_fat),2) AS avg_trans_fat,
    ROUND(AVG(cholesterol),2) AS avg_cholesterol,
    ROUND(AVG(sodium),2) AS avg_sodium,
    ROUND(AVG(total_carb),2) AS avg_carbohydrates,
    ROUND(AVG(fiber),2) AS avg_fiber,
    ROUND(AVG(sugar),2) AS avg_sugar,
    ROUND(AVG(protein),2) AS avg_protein
FROM food
WHERE restaurant = 'Taco Bell';
```

Identification of Highest & lowest Calorie Items:

To identify the ten highest & lowest calorie menu-item records across the dataset.

```sql
#Highest-calorie Taco Bell items
SELECT
    item,
    calories
FROM food
WHERE restaurant = 'Taco Bell'
ORDER BY calories DESC
LIMIT 10;

#Lowest-calorie Taco Bell items
SELECT
    item,
    calories
FROM food
WHERE restaurant = 'Taco Bell'
ORDER BY calories ASC
LIMIT 10;
```

Data Quality and Duplicate Checks:

To identify restaurant-item combinations appearing more than once. Two Taco Bell item names were previously identified as repeated: Chili Cheese Burrito and Express Taco Salad w/ Chips. These should be checked against the source data to determine whether they are genuine duplicate records or separate entries with identical names.

```sql
SELECT restaurant, item, COUNT(*) AS duplicate_count
FROM food
GROUP BY restaurant, item
HAVING COUNT(*) > 1;

SELECT *
FROM food
WHERE restaurant = 'Taco Bell'
AND item IN (
    'Chili Cheese Burrito',
    'Express Taco Salad w/ Chips'
);

SELECT DISTINCT *
FROM food;
```

Summary:

SQL supported the investigation of restaurant-level menu counts, average nutritional values, menu-item rankings, and data quality. The results were subsequently used to support the Excel analysis and Power BI dashboard.

The SQL analysis provided a structured foundation for evaluating Taco Bell's nutritional characteristics and comparing its menu items with those of other restaurants.

Diagnostic Analysis of Taco Bell

Restaurant Performance Analysis:

The restaurant performance analysis compares the nutritional characteristics of eight fast food restaurants using average calories, protein, total fat, sugar, and sodium. The objective is to identify differences in nutritional composition and understand which restaurants offer comparatively higher- or lower-nutrient menu options.

Average Calories by Restaurant:

The overall average calorie content across the analysed menu items is 531.11 calories per item. McDonald's has the highest average calorie content, at approximately 640 calories per item, followed by Sonic at approximately 632 calories and Burger King at approximately 609 calories. Taco Bell has an average of 443.01 calories per item.

Diagnostic interpretation: The higher averages at McDonald's, Sonic, and Burger King suggest that their menus contain a greater average energy content per item. However, the averages alone do not establish the specific ingredients or preparation methods responsible for these differences.

Recommendation: Restaurants with higher average calorie values should review their higher-calorie offerings and consider providing more lower-calorie alternatives. Nutritional information should be clearly displayed to help customers make informed choices.

Average Protein by Restaurant:

The overall average protein content is 27.92 grams per menu item. Based on the dashboard comparison, McDonald's has the highest average protein content, while Taco Bell has a comparatively lower average of approximately 17.38 grams.

Diagnostic interpretation: Differences in average protein may reflect variations in the types and nutritional composition of menu items offered by each restaurant. A higher average protein value does not necessarily mean that every item in that restaurant is protein-rich or nutritionally balanced.

Average Total Fat by Restaurant:

The overall average total fat content is 26.60 grams per menu item. Sonic has the highest average total fat content, at approximately 38 grams per item, followed by Burger King at approximately 37 grams. Chick-fil-A has the lowest average total fat content, at approximately 16 grams.

Diagnostic interpretation: The differences suggest that the fat content of menu items varies across restaurants. Restaurants with higher average total fat may have a greater concentration of high-fat items in their menus. However, total fat alone does not provide a complete assessment of nutritional quality; saturated and trans fat should also be considered.

Average Sugar by Restaurant:

The overall average sugar content for Taco Bell is 3.69 grams per menu item. The highest- and lowest-sugar restaurants across all eight chains should be identified using the Power BI average sugar by restaurant visual or the corresponding SQL results.

Diagnostic interpretation: Sugar levels may vary according to the types of items included in each restaurant's menu. Restaurant-wide averages should be interpreted carefully because individual menu items can differ substantially from the overall average.

Menu Size Analysis:

The menu size analysis compares the number of menu items available at each restaurant. The dashboard reports eight restaurants and 503 menu items in the analysed dataset.

Findings

- Taco Bell has the largest number of menu items, with 113 items.
- Subway has 96 items, followed by Burger King with 70 items.
- McDonald's has 57 items, Arby's has 55, and Sonic has 53.
- Dairy Queen has 42 items.
- Chick-fil-A has the smallest number of menu items, with 27 items.

Diagnostic interpretation: The difference between Taco Bell's 113 items and Chick-fil-A's 27 items indicates substantial variation in menu size across the restaurants. Taco Bell has more than four times as many listed items as Chick-fil-A in this dataset.

A larger menu provides customers with more listed choices, but it does not necessarily indicate better nutritional quality. Similarly, a smaller menu does not automatically mean that a restaurant offers healthier options.

Nutritional Relationship Analysis:

Relationship Between Calories and Protein

The scatter plot comparing calories and protein is used to investigate whether menu items with higher calorie values also tend to contain more protein.

Finding: The scatter plot shows a general upward pattern, suggesting a positive relationship between calories and protein for at least some menu items. The points are widely dispersed, indicating that the relationship is not uniform across all items.

Diagnostic interpretation: The positive pattern suggests that some higher-calorie menu items also contain more protein. However, the spread of points indicates that calorie content cannot be explained by protein alone. Items with similar protein values may have different calorie levels because of differences in fat, carbohydrates, portion size, and other ingredients.

Some menu items appear at substantially higher calorie values than most observations. These items should be examined individually to determine whether their calorie levels are also associated with higher fat or carbohydrate content.

Identification of High-Calorie Items:

The Taco Bell dashboard identifies several items with high calorie values, including items beginning with names such as XXL Grilled and Steak Quesadilla. These examples should be interpreted using their complete menu item names and the corresponding calorie values in the dataset.

Diagnostic interpretation: High-calorie items can contribute to a restaurant's overall calorie profile, particularly when their values are substantially above the restaurant's average. Nevertheless, the presence of a few high-calorie items does not by itself prove that an entire restaurant has a high average calorie content.

Nutritional Differences Between Restaurants:

The comparison of average calories, protein, and total fat shows that restaurants differ in their overall nutritional profiles. McDonald's has a comparatively high average calorie and protein content, while Sonic has a high average total fat content. Taco Bell's average calorie content is lower than those of McDonald's, Sonic, and Burger King.

These findings suggest that restaurant-level nutritional averages can help identify areas for further investigation. However, averages should be supported by individual-item analysis because menu items within the same restaurant may have very different nutritional values.

Conclusion:

The diagnostic analysis demonstrates that the eight fast-food restaurants differ in menu size and nutritional composition. Taco Bell has the largest menu in the analysed dataset, while Chick-fil-A has the smallest. McDonald's, Sonic, and Burger King have comparatively high average calorie values, and Sonic and Burger King have comparatively high average total fat values.

The relationship analysis also indicates that higher calorie content may accompany higher protein content for some menu items, although this pattern is not consistent across all observations. Therefore, calories, protein, fat, sugar, and sodium should be considered together when evaluating nutritional quality.

Overall, the findings demonstrate the value of combining SQL for data analysis, Excel for descriptive statistics, and Power BI for interactive visualisation. The results can help restaurants identify menu items for further review, improve nutritional transparency, and evaluate opportunities to provide a broader range of balanced food choices.

Key Findings

1. Menu size varies considerably: Taco Bell has the largest number of menu items in the analysed dataset, while Chick-fil-A has the smallest number.
2. Average calorie content differs across restaurants: McDonald's, Sonic, and Burger King have comparatively high average calorie values. Taco Bell's average is 443.01 calories per item.
3. Protein content varies: McDonald's has the highest average protein value visible in the restaurant comparison, while Taco Bell has a comparatively lower average.
4. Fat content differs between restaurants: Sonic and Burger King have comparatively high average total fat values, whereas Chick-fil-A has a lower average.
5. Nutritional characteristics are related: The calories-versus-protein scatter plot suggests a positive relationship for some items, but substantial variation remains.
6. Individual menu items require attention: Restaurant averages can conceal individual items with particularly high calories, fat, or other nutrients.
7. Further comparisons are necessary: Verified sugar, sodium, and salad-category counts are required to complete the corresponding restaurant rankings and category analysis.:

Recommendations

Based on the results of the analysis:

1. Improve nutritional transparency
   Restaurants should provide clear and consistent nutritional information for menu items, including calories, protein, total fat, sugar, and sodium. This would allow customers to compare options more effectively.
2. Review high-calorie menu items
   Restaurants with higher average calorie values should investigate the individual items contributing to those averages. Lower-calorie alternatives and suitable portion sizes could be considered where feasible.
3. Evaluate high-fat items
   Restaurants with higher average total fat should review their highest-fat offerings, including their saturated and trans fat content. Potential ingredient or preparation improvements should be evaluated without compromising food quality.
4. Expand balanced menu options
   Restaurants should assess opportunities to provide items that combine suitable protein content with moderate calories, fat, and sodium. Nutritional quality should be evaluated across multiple measures rather than by protein content alone.
5. Monitor sodium and sugar levels
   Once the restaurant rankings are confirmed, businesses with higher average sodium or sugar should investigate the menu items contributing most to those values and assess opportunities for improvement.

Conclusion

The Fast Food Analysis project used SQL, Excel, and Power BI to investigate nutritional characteristics across fast-food restaurants, with Taco Bell selected for detailed analysis.

The project demonstrated how data analysis tools can be used to summarize nutritional information, identify menu items with extreme nutritional values, and present findings through visualizations.

The resulting analysis provides a structured view of Taco Bell's menu and supports nutritional comparisons using the available dataset.